

Farhan Fuad Abir

(+1) 689-270-1830
farhan.fuad@ucf.edu
Orlando, FL, US

github.com/farhanfuadabir
[linkedin.com/in/farhanfuadabir](https://www.linkedin.com/in/farhanfuadabir)

ACADEMIC QUALIFICATIONS **University of Central Florida**, Florida, US Aug 2024 - Present
Doctor of Philosophy in Computer Engineering

Supervisor: Dr. Laura Brattain

Area: Working on ultrasound-based disease diagnosis with multimodal AI, exploring trustworthiness with explainable AI and exploring generative AI-based solutions to class imbalance

University of Central Florida, Florida, US

Aug 2022 - Aug 2024

Master of Science in Computer Engineering

CGPA: 3.89/4.00

Supervisor: Dr. Rickard Ewetz

Thesis: [Neuro-Symbolic Distillation of Reinforcement Learning Agents](#)

- Developed an LLM-assisted neuro-symbolic feature extraction method for visual observation space

- Proposed a neural-guided symbolic model extraction method for continuous observation space

University of Dhaka, Dhaka, Bangladesh

2016 - 2020

Bachelor of Science in Electrical and Electronic Engineering

CGPA: 3.55/4.00

Supervisor: Dr. Mosabber Uddin Ahmed

Project: [A Framework for Sensor-Based Sign Language Detection using Machine Learning](#)

- Developed a low-cost dataglove to detect 14 static and 3 dynamic hand gestures in real-time

- Made two datasets of static and dynamic gestures of 35 subjects using flex and IMU sensors

SELECTED PUBLICATIONS

Conference Proceedings

- Mahad Ali, **F. F. Abir**, T. Overton, M. Rakhshan, L. E. Perotti, & L. J. Brattain, Real-time Median Nerve and Vessel Tracking in Ultrasound on Resource-constrained Devices, in *48th IEEE Engineering in Medicine and Biology Society (EMBC)*, 2026
- **F. F. Abir**, S. S. Jennifer, N. Yousefi, & L. J. Brattain, "[Hybrid Diffusion Model for Breast Ultrasound Image Augmentation](#)", in *2026 IEEE International Symposium on Biomedical Imaging (ISBI)*, 2026.
- **F. F. Abir**, A. E. Daly, K. Anderman, T. Ozmen, & L. J. Brattain, "[Multimodal Deep Learning for Phyllodes Tumor Classification from Ultrasound and Clinical Data](#)", in *21st IEEE BSN'25*, 2025.
- M. A. A. Faisal, **F. F. Abir**, M. U. Ahmed, "[Sensor Dataglove for Real-time Static and Dynamic Hand Gesture Recognition](#)", in *10th ICIEV'21*, 2021, pp. 1-7. **[Best Paper]**
- M. S. Siraj, M. A. A. Faisal, O. Shahid, **F. F. Abir**, T. Hossain, S. Inoue, and M. A. R. Ahad, "[UPIC: User and position independent classical approach for locomotion and transportation modes recognition](#)", in *Adjunct Proceedings of the UbiComp-ISWC'20*, 2020, pp. 340-345.
- M. A. A. Faisal, M. S. Siraj, M. T. Abdullah, O. Shahid, **F. F. Abir**, and M. Ahad, "[A pragmatic signal processing approach for nurse care activity recognition using classical machine learning](#)", in *Adjunct Proceedings of the UbiComp-ISWC'20*, 2020, pp. 396-401.

Journal Articles

- **F. F. Abir**, M. E. H. Chowdhury, M. I. Tapotee, et al., "[PCovNet+: A CNN-VAE anomaly detection framework with LSTM embeddings for smartwatch-based COVID-19 detection](#)", in *Engineering Applications of Artificial Intelligence* [IF - 7.802 (2023)], 2023, 106130
- S. Mahmud, N. Ibtehaz, A. J. Gonzales, T. Rahman, M. S. Hossain, M. S. A. Hossain M. A. A. Faisal, **F. F. Abir**, et al., "[NABNet: A Nested Attention-guided BiConvLSTM network for a robust prediction of Blood Pressure components from reconstructed Arterial Blood Pressure waveforms using PPG and ECG signals](#)", in *Biomedical Signal Processing and Control* [IF - 5.076 (2023)], 2022, 104247
- M. N. I. Shuzan, M. E. H. Chowdhury, M. B. I. Reaz, A. Khandakar, **F. F. Abir**, et al., "[Machine learning-based classification of healthy and impaired gaits using 3D-GRF signals](#)", in *Biomedical Signal Processing and Control* [IF - 5.076 (2023)], 2023, 104448

- **F. F. Abir**, K. Alyafei, M.E.H. Chowdhury, A. Khandakar, R. Ahmed, et al. "[PCovNet: A presymptomatic COVID-19 detection framework using deep learning model using wearables data](#)", in *Computers in Biology and Medicine*, 2022, 105682 [IF - 6.69 (2022)]
- M. A. A. Faisal, **F. F. Abir**, M. U. Ahmed, M. E. H. Chowdhury, "Exploiting domain transformation and deep learning for hand gesture recognition using a low-cost dataglove", in *Scientific Reports* [IF - 4.997 (2022)], 2022, 21446
- K. Alyafei, R. Ahmed, **F. F. Abir**, M. E. H. Chowdhury, et al., "A comprehensive review of COVID-19 detection techniques: From laboratory systems to wearable devices", in *Computers in Biology and Medicine* [IF - 6.698 (2022)], 2022, 106070
- T. Rahman, A. Khandakar, **F. F. Abir**, M. A. A. Faisal, M. S. Hossain, et al., "[QCovSML: A reliable COVID-19 detection system using CBC biomarkers by a stacking machine learning model](#)", in *Computers in Biology and Medicine*, 2022, 105284 [IF - 6.69 (2022)]
- M. M. Soliman, M. E. H. Chowdhury, M. T. Islam, F. Musharavati, M. Nabil, M. Hafizh, A. Khandakar, S. Mahmud, E. Z. Nezhad, M. N. I. Shuzan, **F. F. Abir**, "A Review of Biomaterials and Associated Performance Metrics Analysis in Pre-Clinical Finite Element Model and in Implementation Stages for Total Hip Implant System", in *Polymers* [IF - 4.967 (2022)], 2022, 14, 4308

Book Chapter

- **F. F. Abir**, M. A. A. Faisal, O. Shahid, M. U. Ahmed, "[Contactless Human Activity Analysis: an overview of different modalities](#)", in *Contactless Human Activity Analysis*, Springer Nature Switzerland AG, 2021, vol. 200, pp. 83-112.

ATTENDED CONFERENCES	IEEE-EMBS International Conference on Body Sensor Networks 2025	Nov 2025
	IEEE CS BDS Winter Symposium'20 [Online]	Nov 2020
	<i>Presentation:</i> Locomotion and Transportation Modes Recognition Using Smart Feature based Machine Learning	
	UbiComp-ISWC'20 [Online]	Sep 2020
	ICIEV-IVPR-ABC'20 [Online]	Aug 2020
INVITED TALKS	PCB Design 101	Aug 2020
	- Designed the 10-days online course curriculum delivered via Microsoft Teams and managed with Google Classroom	
	- Introduced the basic steps of Circuit Schematic and PCB designing with Proteus Design Suite	
	- Taught design techniques to 30 participants hands-on in 3 live sessions	
	Discussion on Preliminary Ideas of Machine Learning , IEEE Student Branch DU	Sep 2019
PROFESSIONAL EXPERIENCES	- Presented the scopes of using ML for Hand Gesture Recognition to 45 sophomore students	
	Introduction to Automation , IEEE RAS DU	Jul 2018
	- Introduced the locally available embedded systems modules for automation to 23 freshman students	
	Graduate Research Assistant , Dr. Laura Brattain	Aug 2024 - Present
	<i>Project:</i> Multimodal deep learning model for ultrasound-based disease diagnosis leveraging clinical data	
	- Used the method for breast cancer and pleural effusion diagnosis	
	- Implemented feature-level fusion for ultrasound image and clinical data	
	- Explored CAM-based explainability approaches for diagnosis validation	
	<i>Project:</i> Hybrid diffusion model for high-fidelity breast ultrasound image generation	
	- Used both text-to-image and image-to-image stable-diffusion models to generate breast ultrasound images of different classes	
	<i>Project:</i> Ultrasound-based Neck-landmark Detection	
	- Implemented YOLOv11 for object detection with class-aware sampling to mitigate class imbalance	
	Graduate Research Assistant , Dr. Rickard Ewetz	Oct 2022 - Aug 2024
	<i>Project:</i> Neurosymbolic Reinforcement Learning	
	- Explored the potential of using neurosymbolic approach for interpretable reinforcement learning agents	
	- Mentored 2 undergraduate students on interpretable reinforcement learning	

	Research Assistant , Qatar University Machine Learning Group Aug 2021 - Aug 2022 <i>Project: Presymptomatic COVID-19 Detection Using Smartwatch Data</i> - Worked with 3 biomarkers derived from smartwatch data to extract underlying diseases - Experimented on deep learning-based anomaly detection models <i>Project: IoT-Enabled Automatic IV Dispenser System for Burn Patients</i> - Designed on 3 subsections - saline drip detector, automated saline flow controller, and urine output monitor systems - Integrated the systems in the IoT backend for real-time monitoring and control	
	Biomedical Engineer , Datasoft Manufacturing & Assembly Inc. Jul 2021 - Sep 2021 <i>Project: Virtual Doctor - A IoT-enabled Telemedicine Platform</i> - Developed three devices to monitor vitals – Heart Rate, Blood Pressure, Oxygen Saturation, and Temperature - Integrated the devices in the telemedicine platform for real-time monitoring - Coordinated a team of six interns	
	Embedded System Engineer , Quanfield Technologies Jun 2016 - Jan 2018 <i>Project: ArcherClock</i> - A timer and scoring system for archery competition - Developed the system protocol, designed the PCB of the whole system, and used Raspberry Pi to control and handshake with the mobile app <i>Project: QuizzoBuzz</i> - A buzzer system for quiz competition - Designed the circuits and PCB of six buzzer terminals and programmed the AVR microcontroller	
STANDARDIZED TEST SCORES	GRE - 320 (Quant: 164, Verbal: 156, AWA: 4.0) Oct 2020 IELTS - 7.5 (Listening: 8.5, Reading: 7.5, Writing: 6.5, Speaking: 7) Oct 2021	
RELEVANT SKILLS	<i>Languages:</i> C/C++, Python (PyTorch, TensorFlow, Keras, SciKit Learn, Django) <i>Web:</i> HTML, CSS, Bootstrap <i>Others:</i> Matlab, Circuit Simulation and Prototyping (LTSpice), PCB Design (Proteus, Eagle), PCB Fabrication	
ADDITIONAL PROJECTS	ASL Dataglove - A sensor-based dataglove to interpret 40 ASL gestures 2021 Tothastu - A blog created with Django 2021 BioAI - AI-based medical assistant application based on Google BERT 2020 Pet Robot - Companion robot capable of following a certain color tag 2018 Line Follower Robot - A robot with path following ability under various challenges 2016	
HONORS & AWARDS	ORCGS Doctoral Fellowship, University of Central Florida, United States Aug 2022 Best Paper, Int'l. Conference on Informatics, Electronics & Vision 2021, Japan Aug 2021 Champion, Satellite Expedition Idea Contest, Brac University, Bangladesh Jan 2021 Runner-up, Idea Contest, 1st National Workshop on BDML, Bangladesh Dec 2020 6 th Position, Sussex-Huawei Locomotion Challenge 2020, Cancún, Mexico Sep 2020 Finalist, APSCO Micro-satellite Idea Contest, Xi'an, China Oct 2019 Runner-up, Matlab Contest, IUT, Dhaka, Bangladesh Sep 2019 Runner-up, Project Showcasing, FabLab DU, Dhaka, Bangladesh Nov 2018 Champion, Line Follower Robot Competition, DUET, Dhaka, Bangladesh Jul 2017	
VOLUNTEER EXPERIENCES	President, Bangladeshi Student Association at UCF 2025 - 2026 Secretary and Treasurer, IEEE Student Branch DU 2019 - 2021 Volunteer, FabLab University of Dhaka 2018 Volunteer, CommunityAction (UN-recognized Social Welfare Organization) 2016 - 2017	